

Kronos Error Analysis Report

ETH 5m | 5000 samples | 2025-07-02 to 2026-02-28

DIRECTION (full-sequence decode)

Predicted tokens: 50.3% | Edge: +0.2pp

GT tokens sanity: 83.7% (should be ~100% if decode preserves dir)

Baseline (majority): 50.1% (up=49.7%, down=50.1%)

S1 (coarse)

Token accuracy: 30.1% | Per-head direction: 31.8%

When correct: top1_prob=0.434, entropy=2.27, eff_N=12

When wrong: top1_prob=0.270, entropy=2.87, eff_N=24

Direction correct even when token wrong: 33.7%

S2 (fine)

Token accuracy: 49.7% | Per-head direction: 23.8%

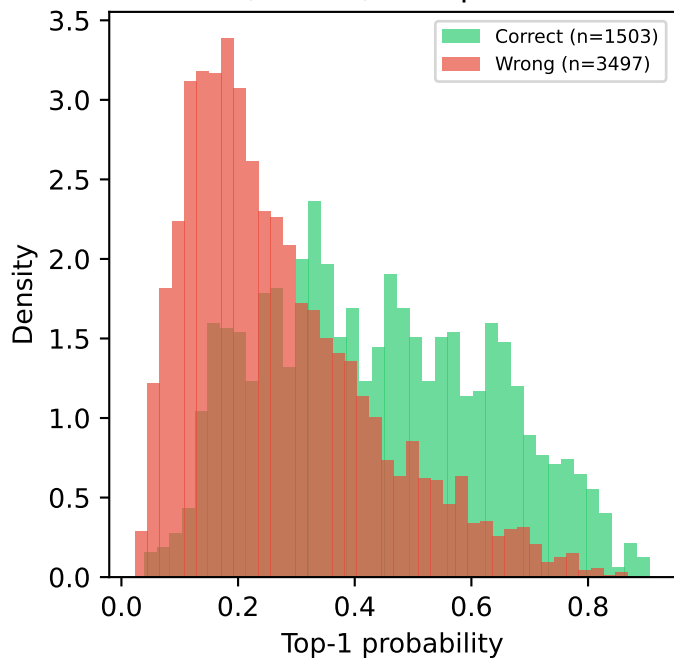
When correct: top1_prob=0.615, entropy=1.47, eff_N=5

When wrong: top1_prob=0.412, entropy=2.01, eff_N=9

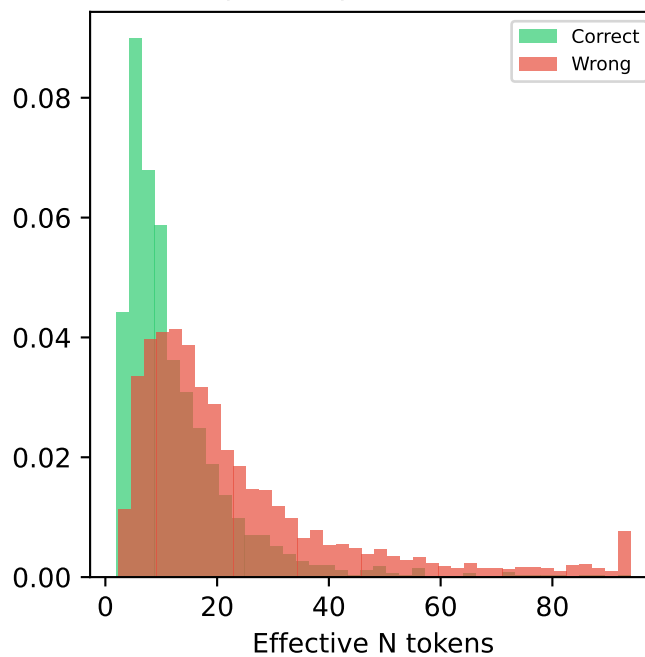
Direction correct even when token wrong: 29.0%

Part 1: When Wrong — Concentrated or Scattered?

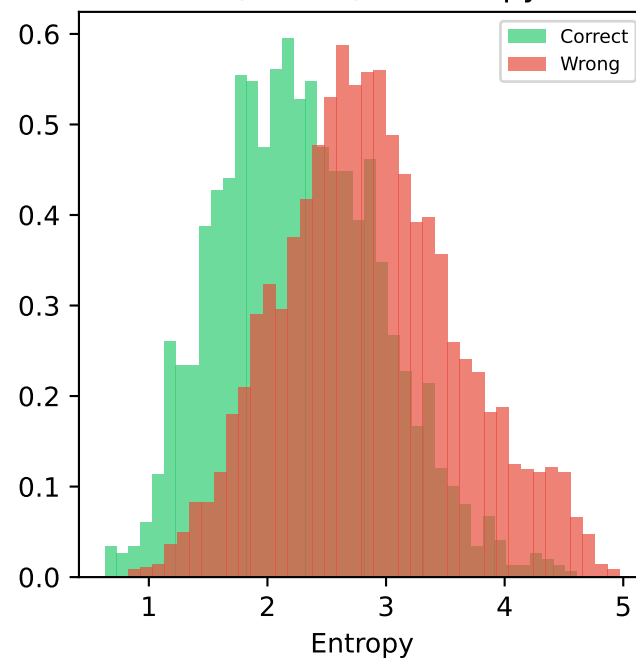
S1 (coarse) — Top-1 Prob



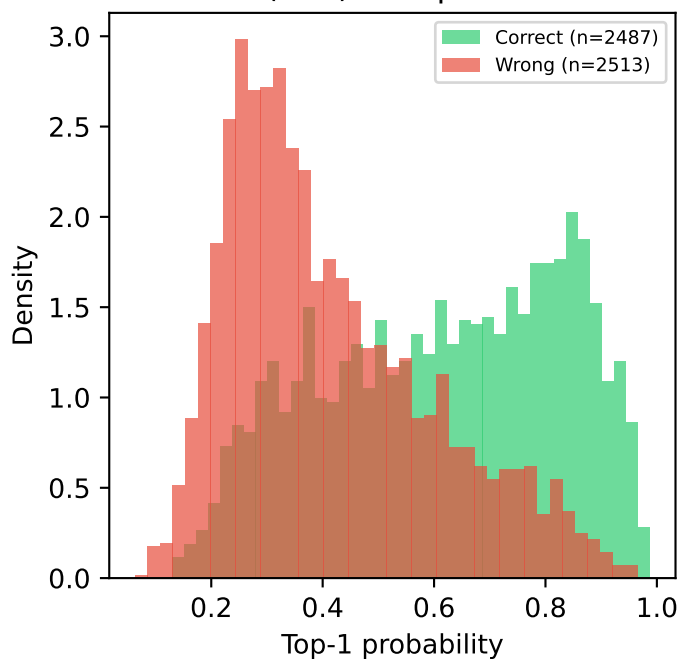
S1 (coarse) — Effective N



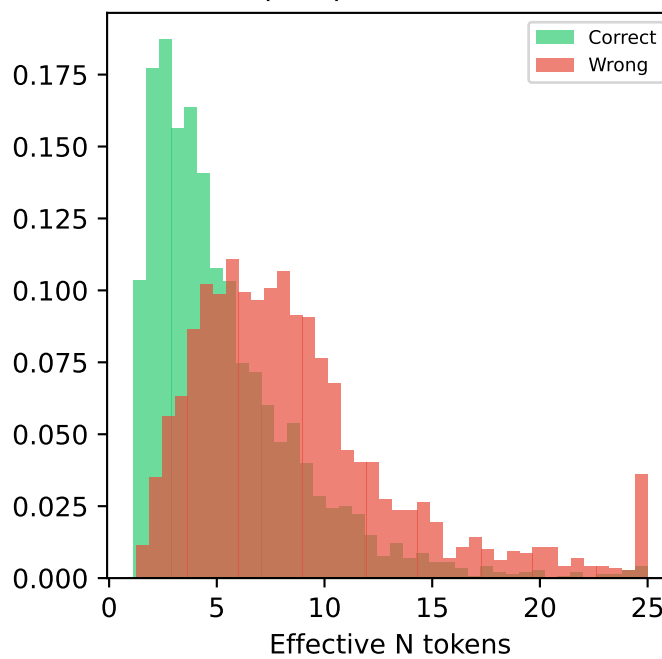
S1 (coarse) — Entropy



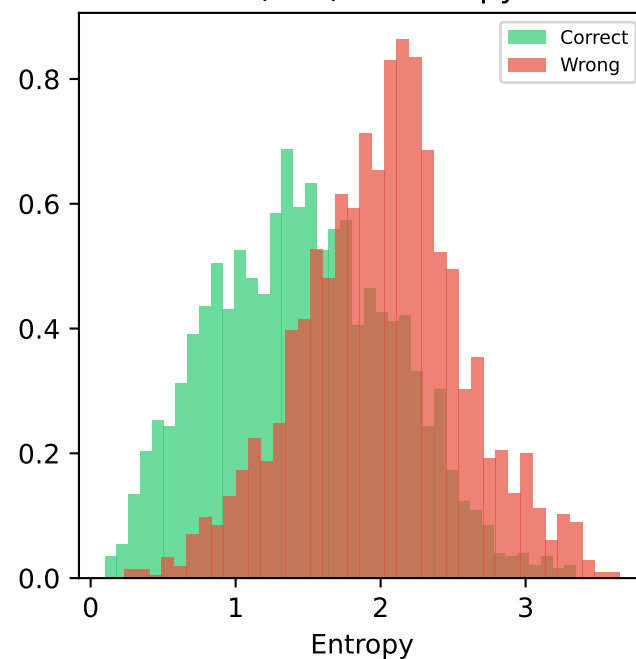
S2 (fine) — Top-1 Prob



S2 (fine) — Effective N

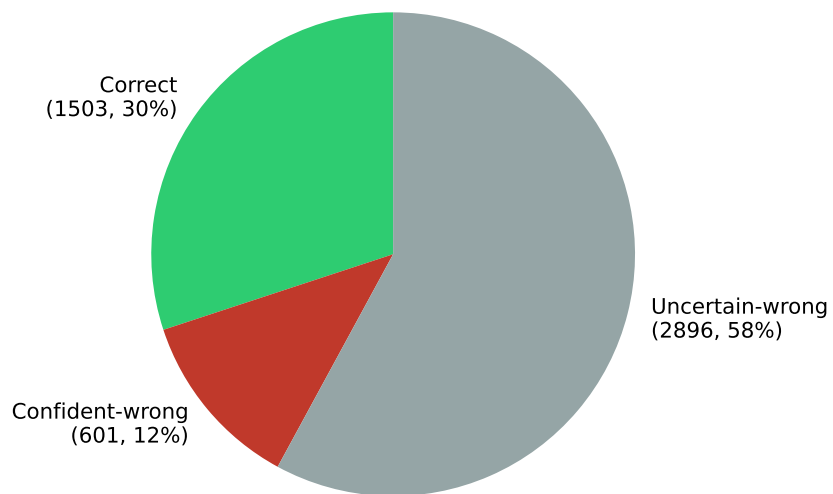


S2 (fine) — Entropy

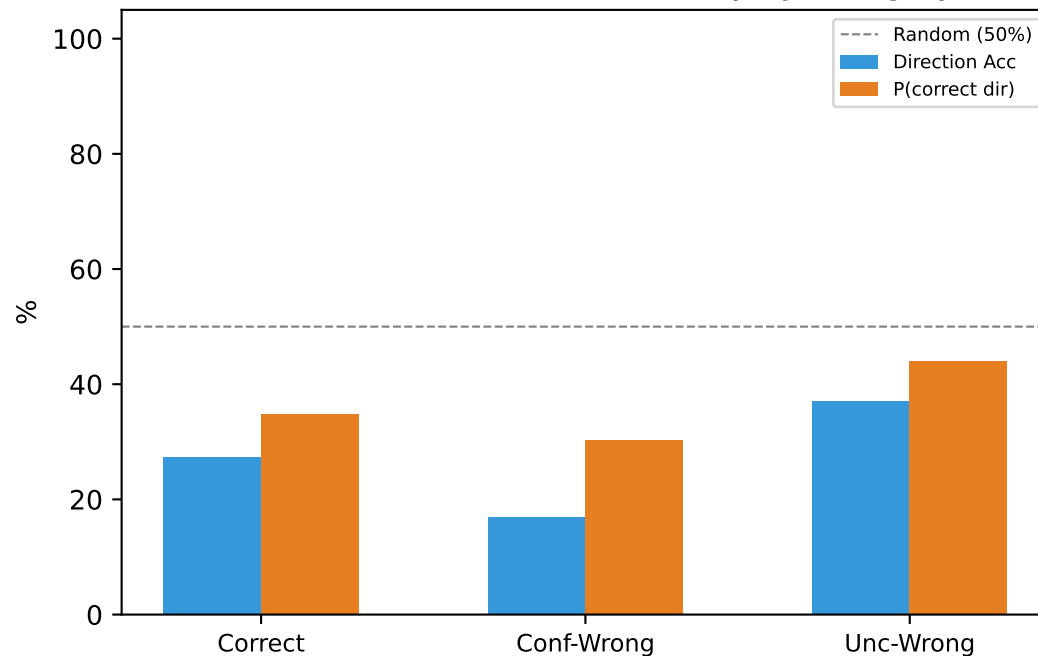


Wrong Predictions Breakdown: Confident vs Uncertain

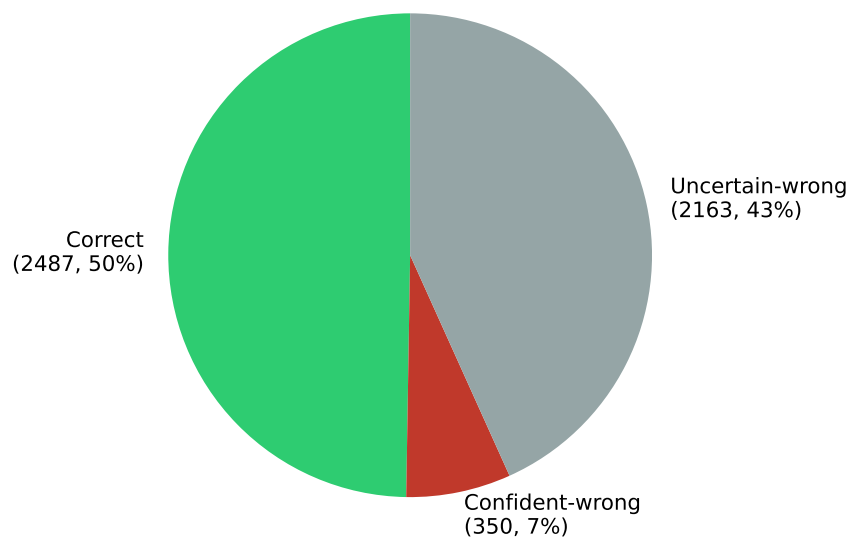
S1 (coarse) — Prediction Categories



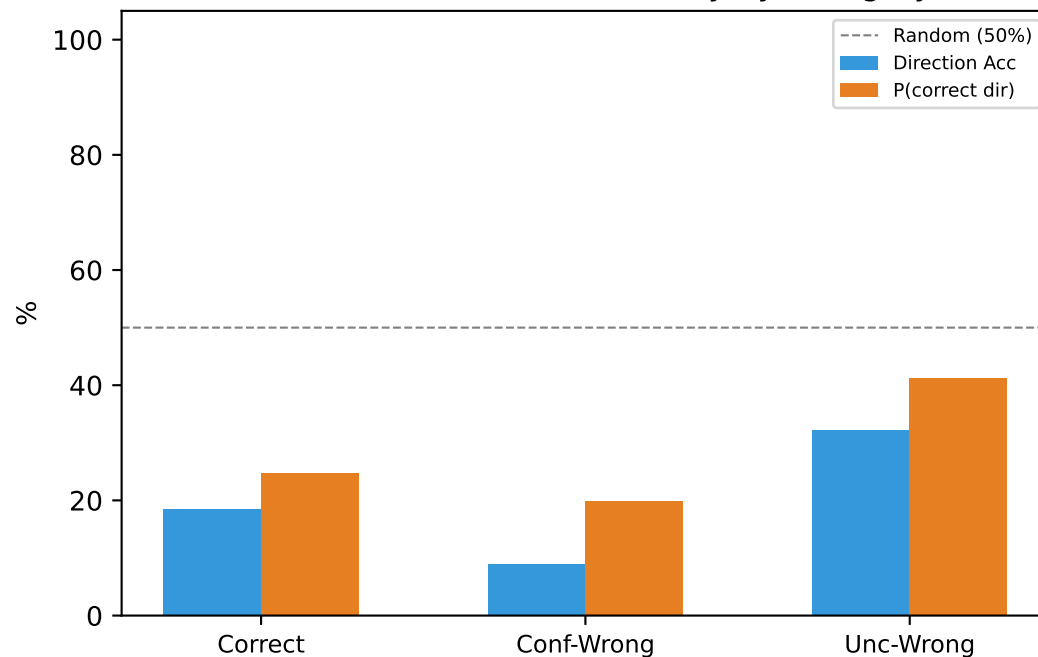
S1 (coarse) — Direction Accuracy by Category



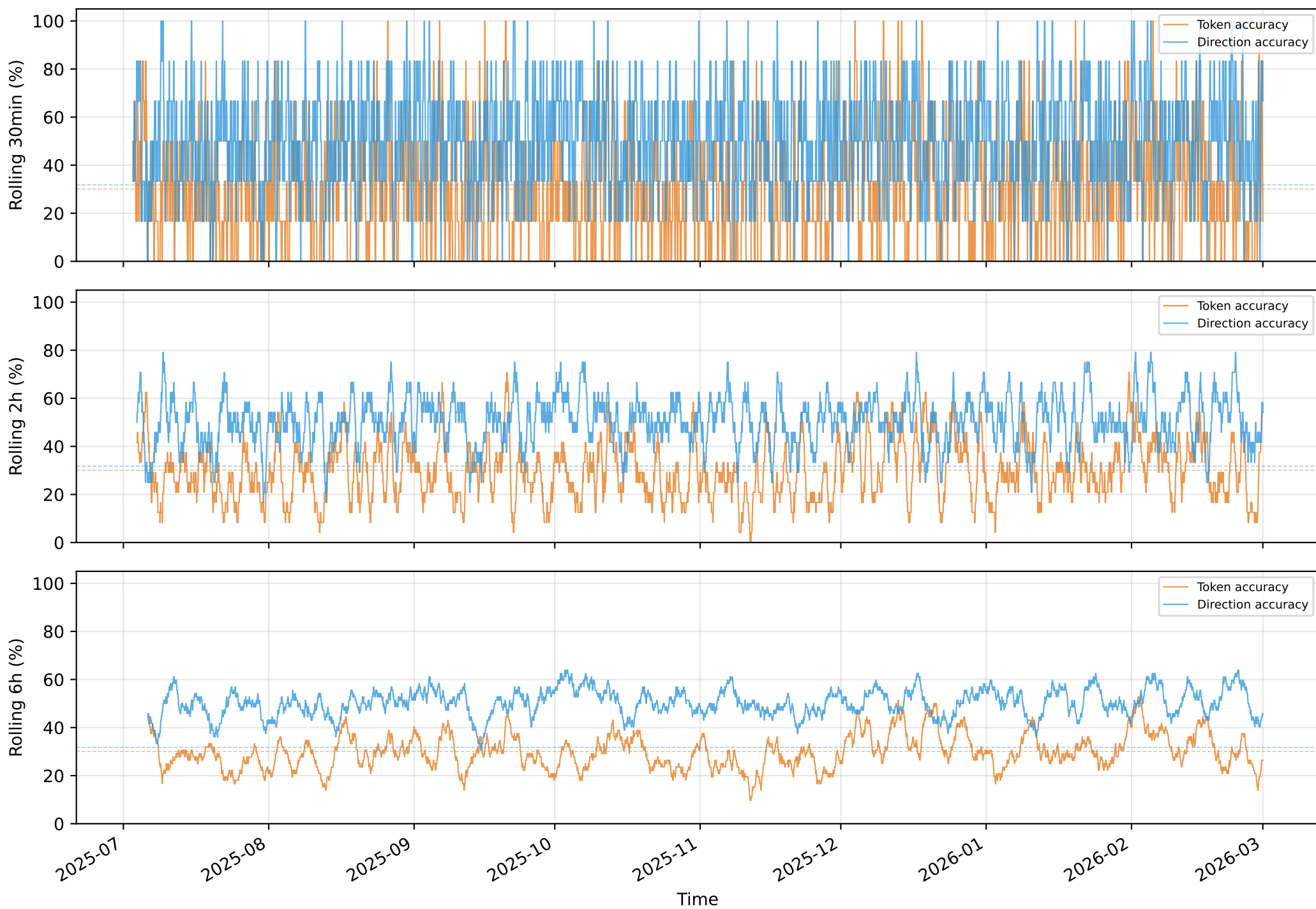
S2 (fine) — Prediction Categories



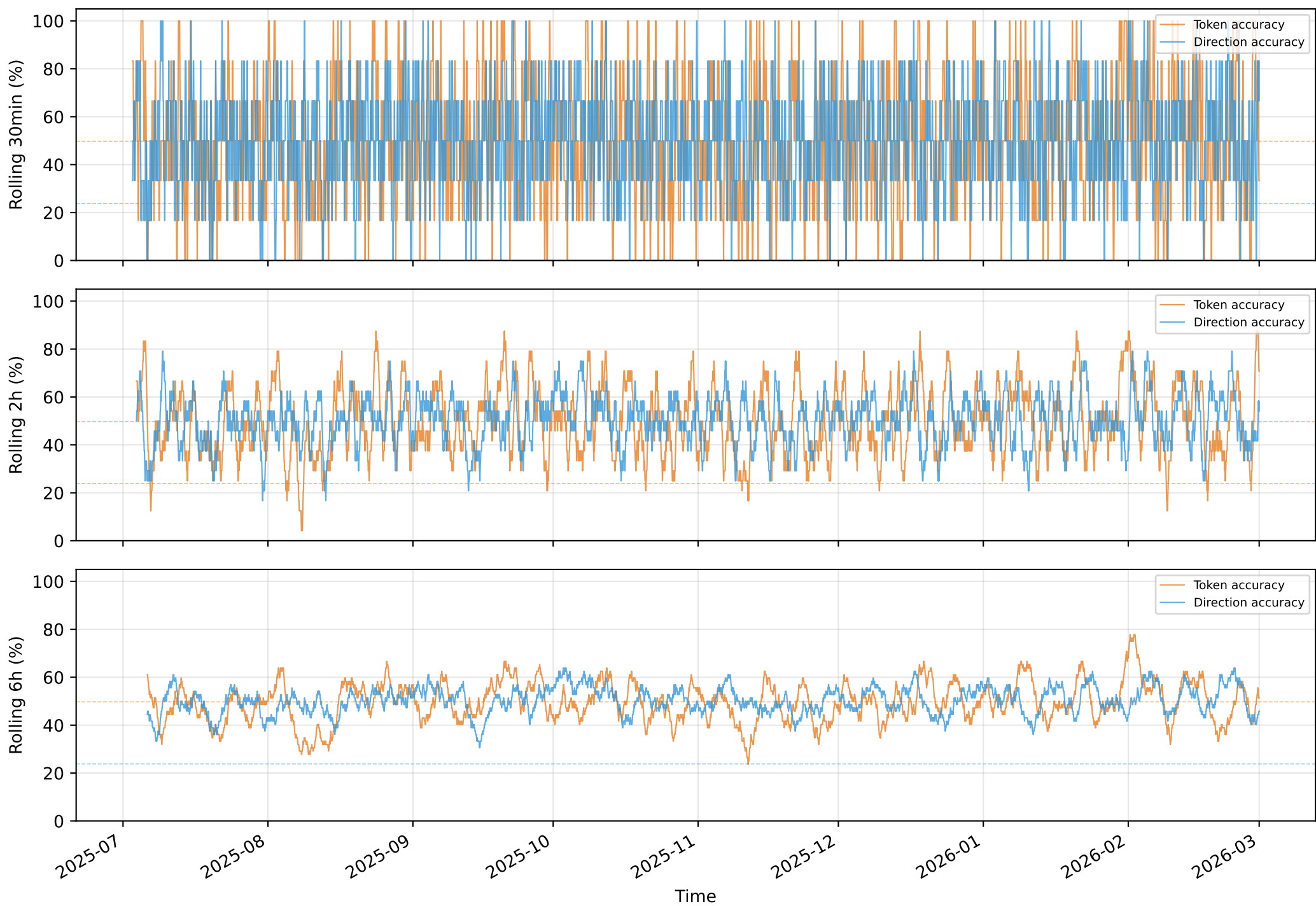
S2 (fine) — Direction Accuracy by Category



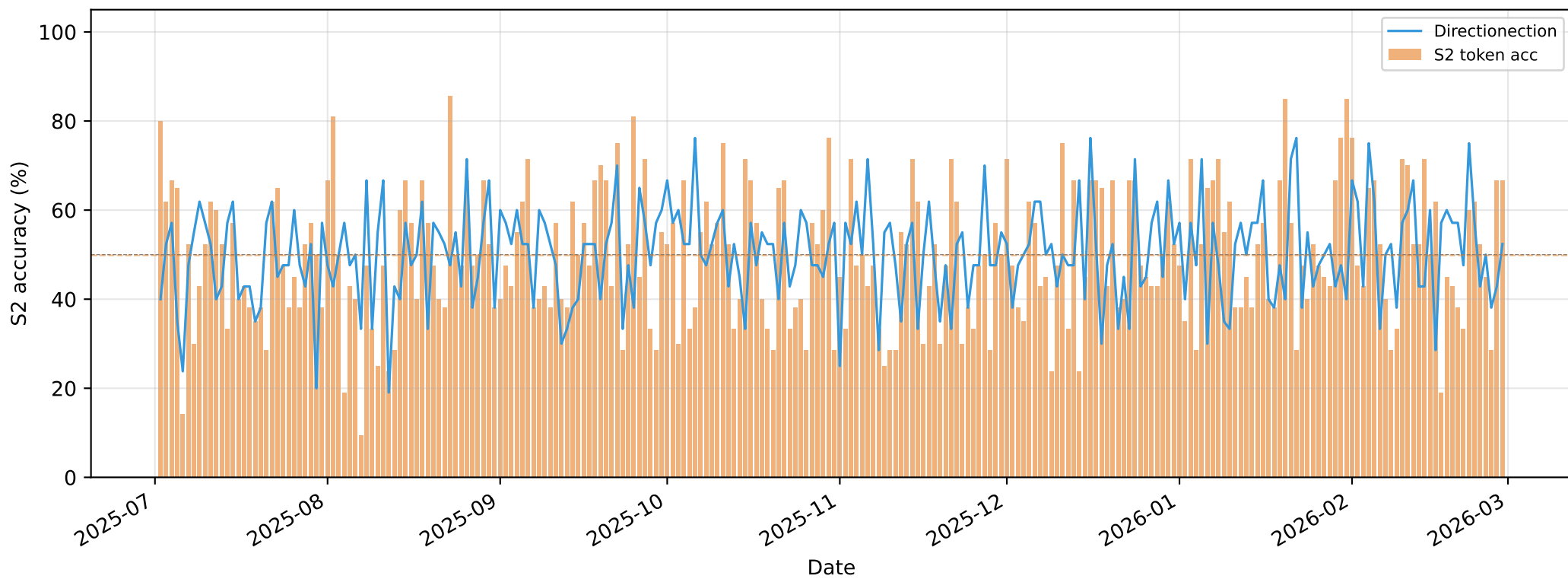
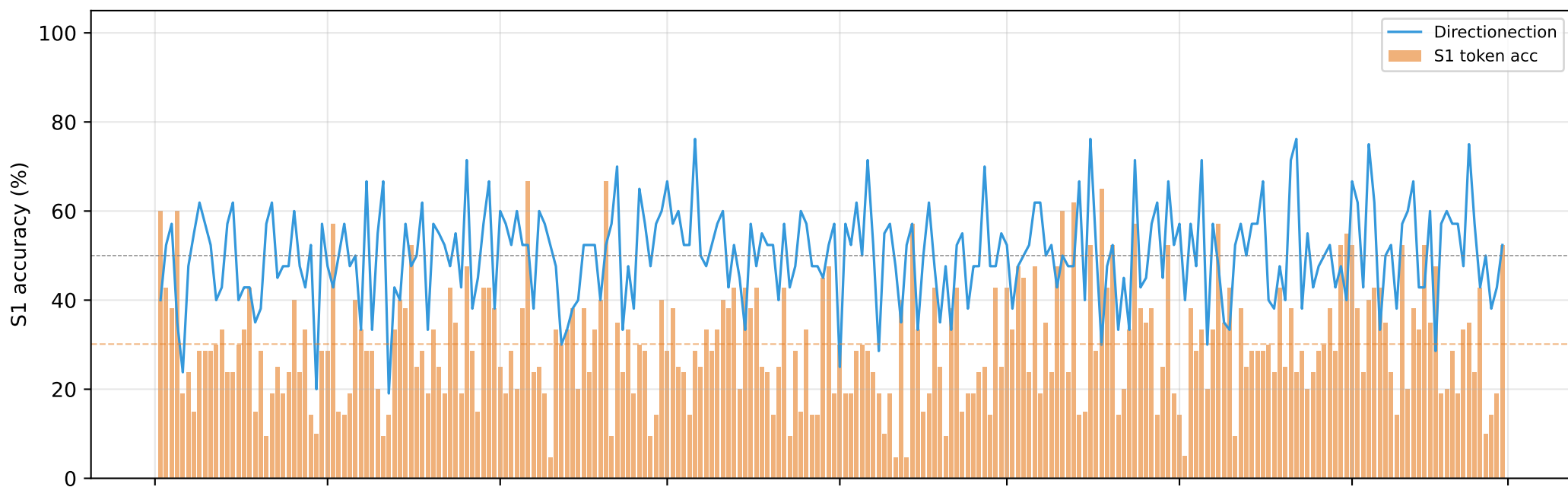
Temporal Analysis — S1 (coarse)



Temporal Analysis — S2 (fine)

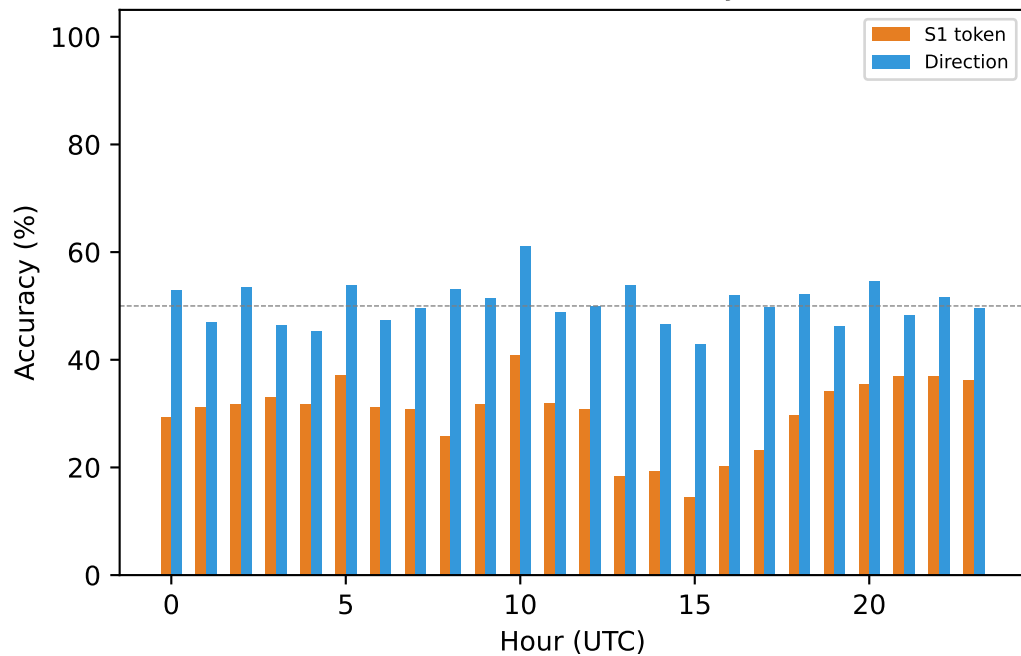


Daily Accuracy (Direction (pred_s1, pred_s2))

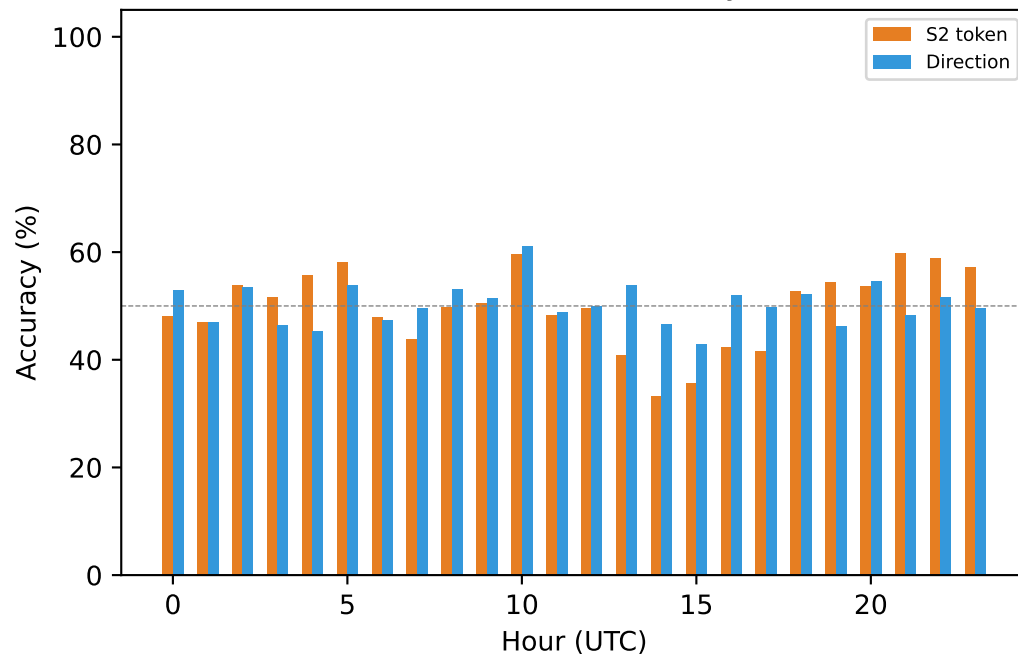


Accuracy by Time of Day

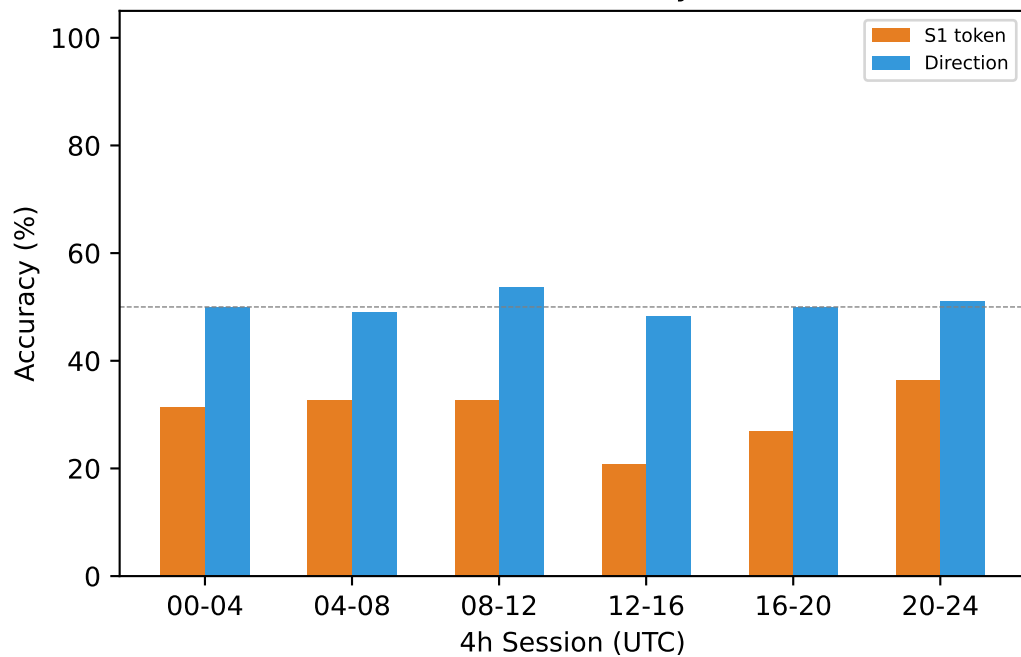
S1 Token + Direction — By Hour



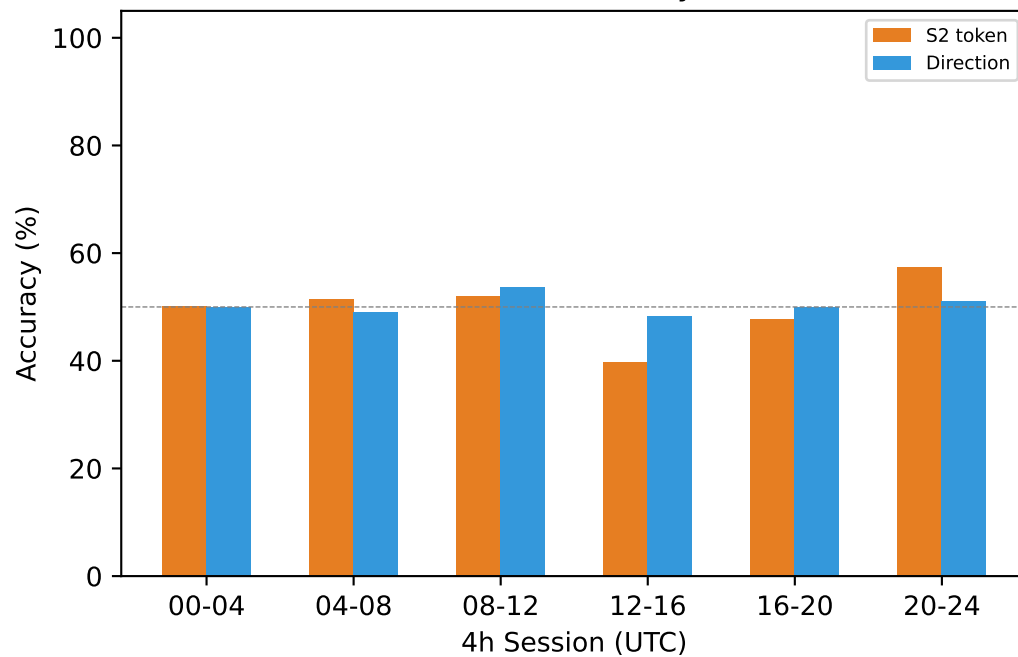
S2 Token + Direction — By Hour



S1 Token + Direction — By 4h Session

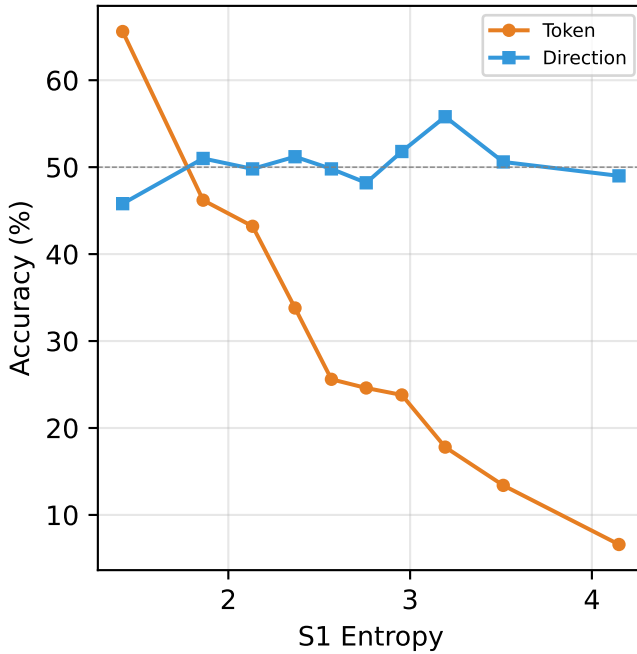


S2 Token + Direction — By 4h Session

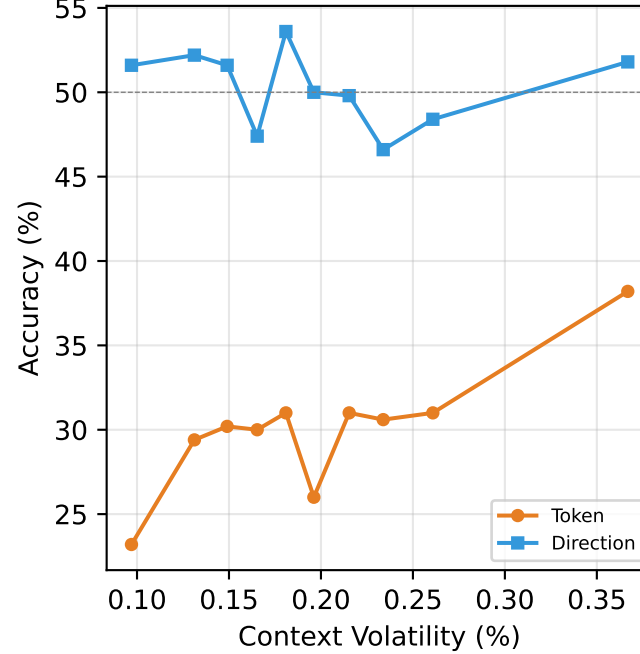


Accuracy vs Market Conditions & Model Confidence

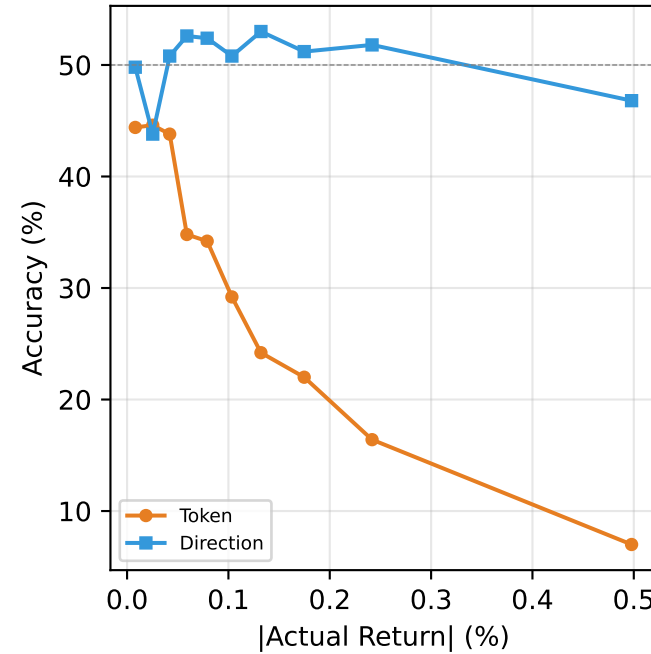
S1 Entropy — vs Accuracy



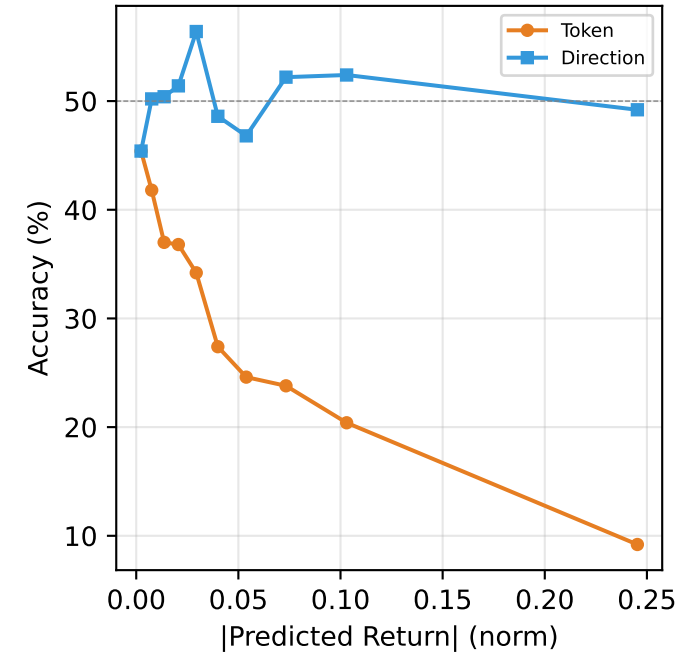
S1 — vs Volatility



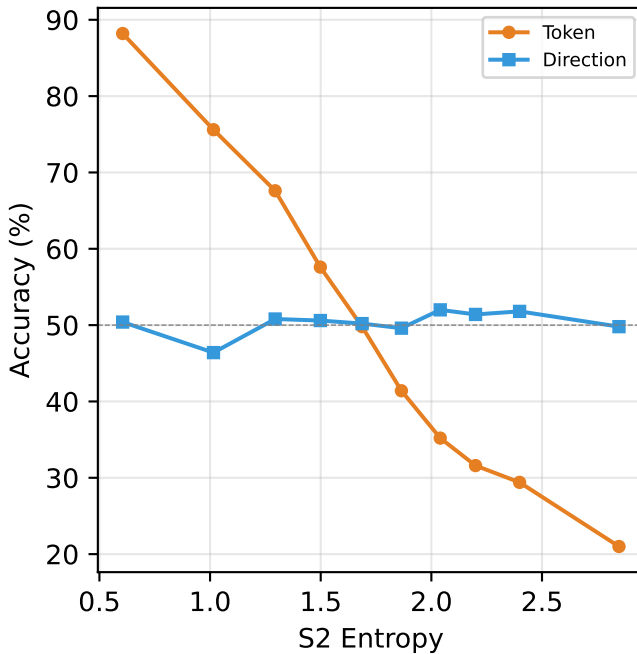
S1 — vs Return Size



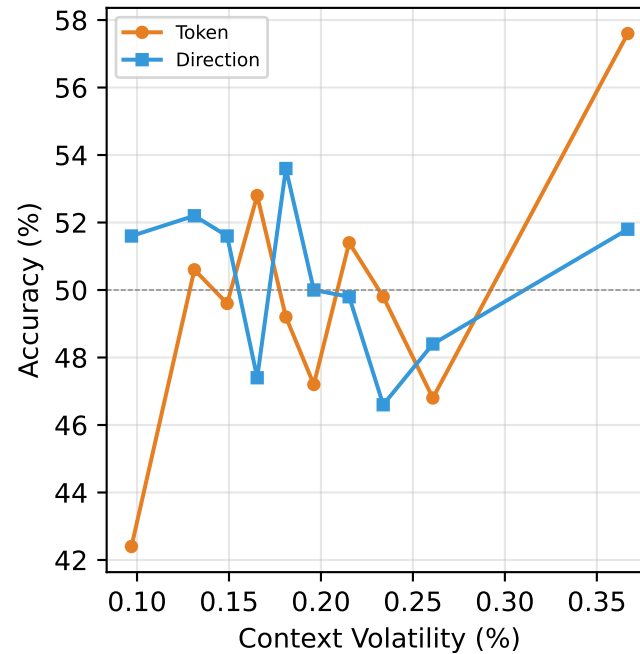
S1 — vs Predicted Move



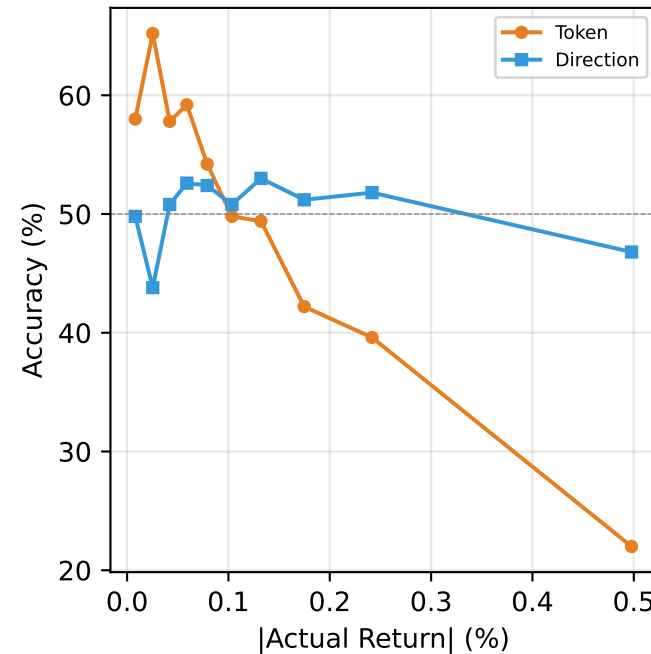
S2 Entropy — vs Accuracy



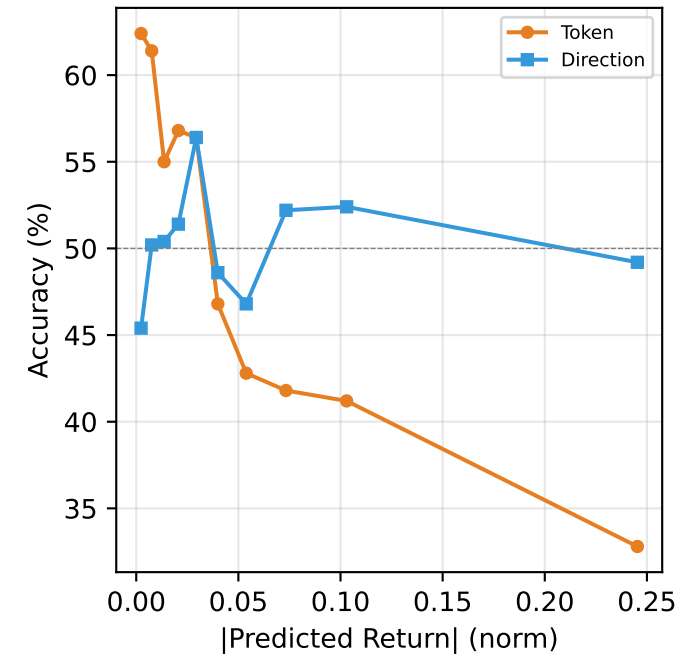
S2 — vs Volatility



S2 — vs Return Size

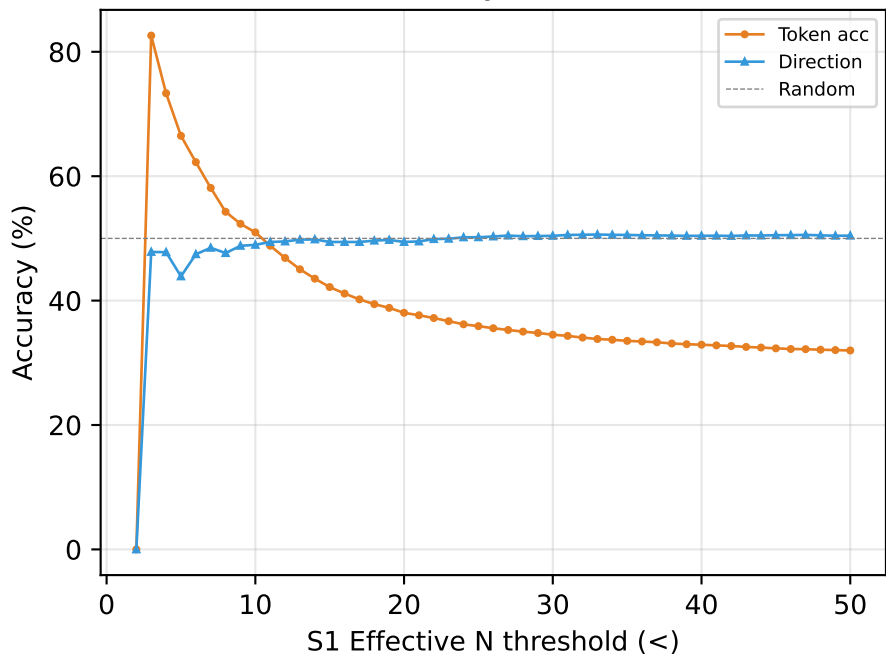


S2 — vs Predicted Move

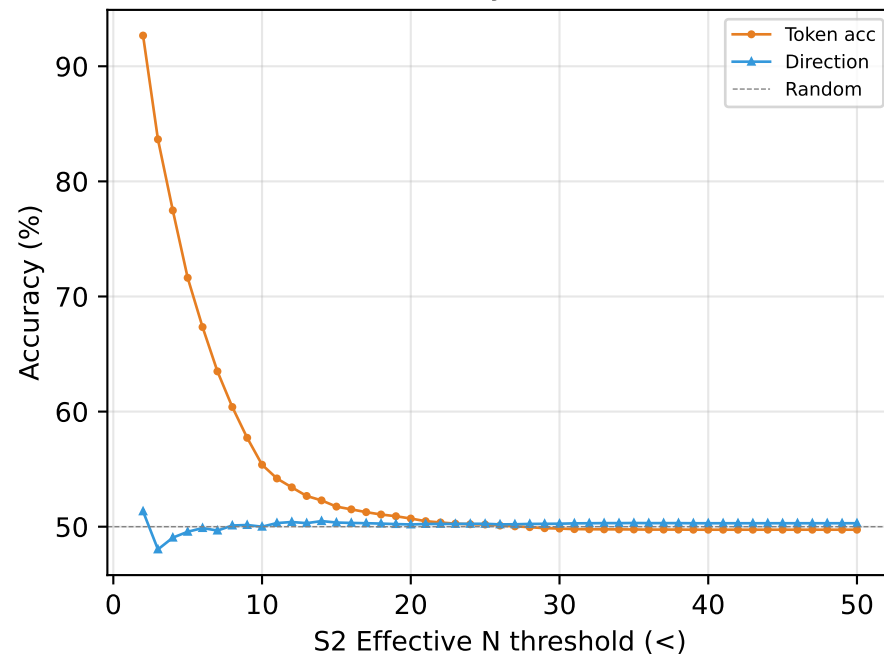


Effective N as Trading Filter — Threshold Sweep

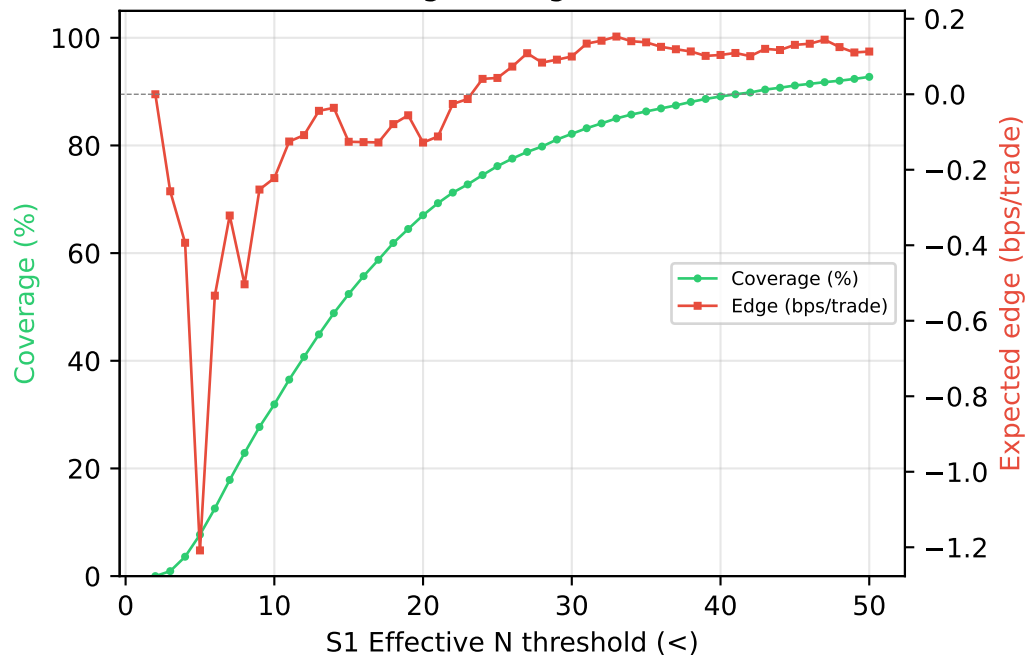
S1 — Accuracy vs Eff N Filter



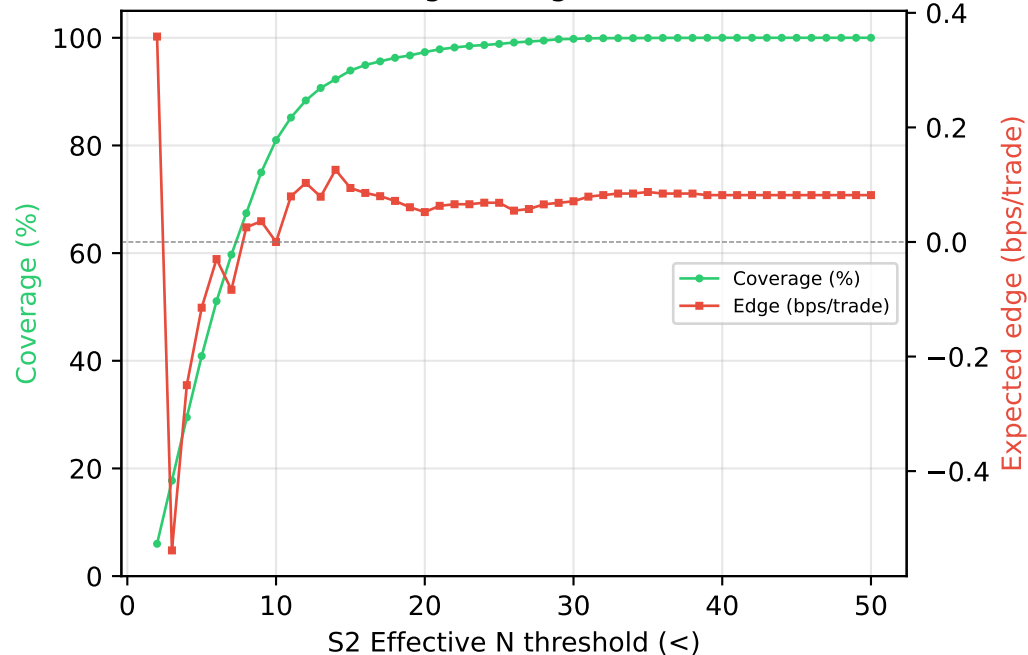
S2 — Accuracy vs Eff N Filter



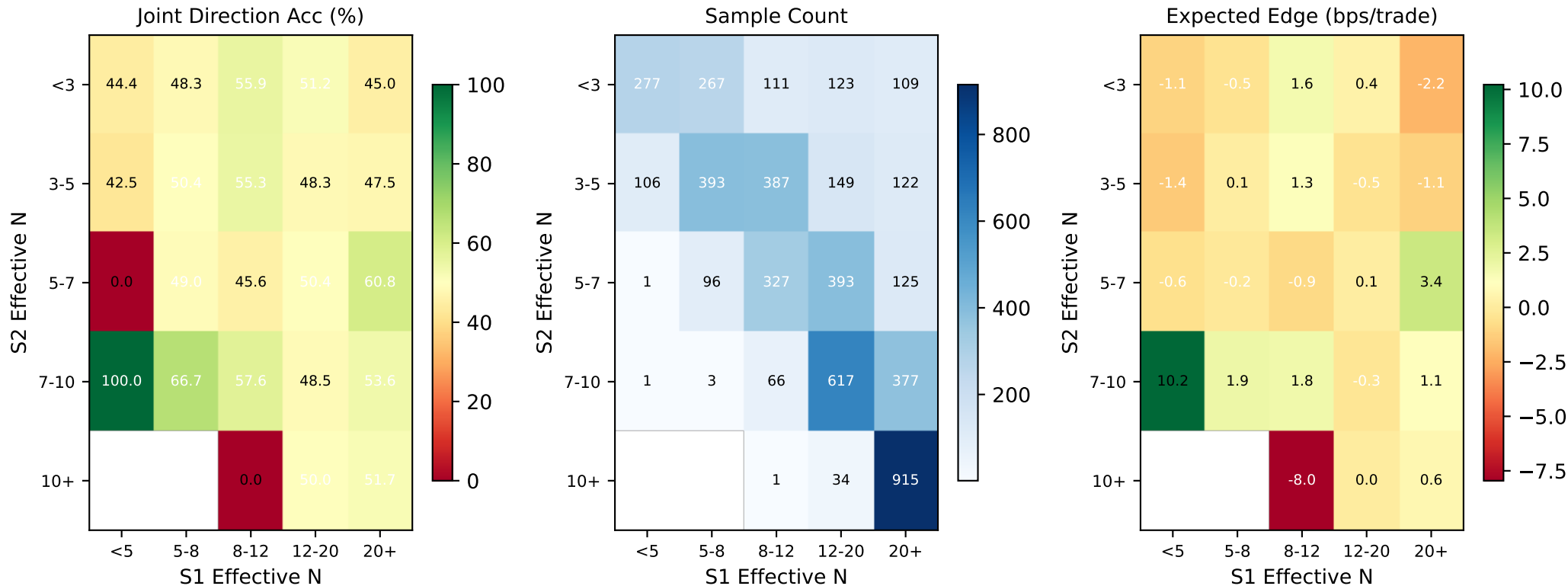
S1 — Coverage & Edge vs Eff N Filter



S2 — Coverage & Edge vs Eff N Filter

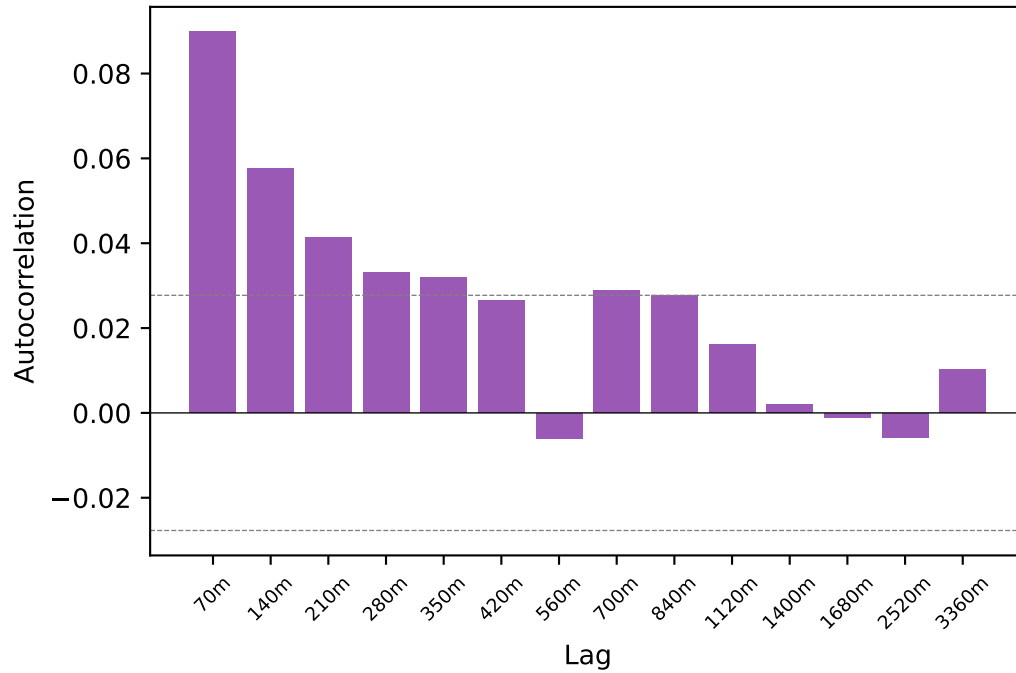


Cross-Head Filter: Direction Accuracy by (S1 eff_n, S2 eff_n)

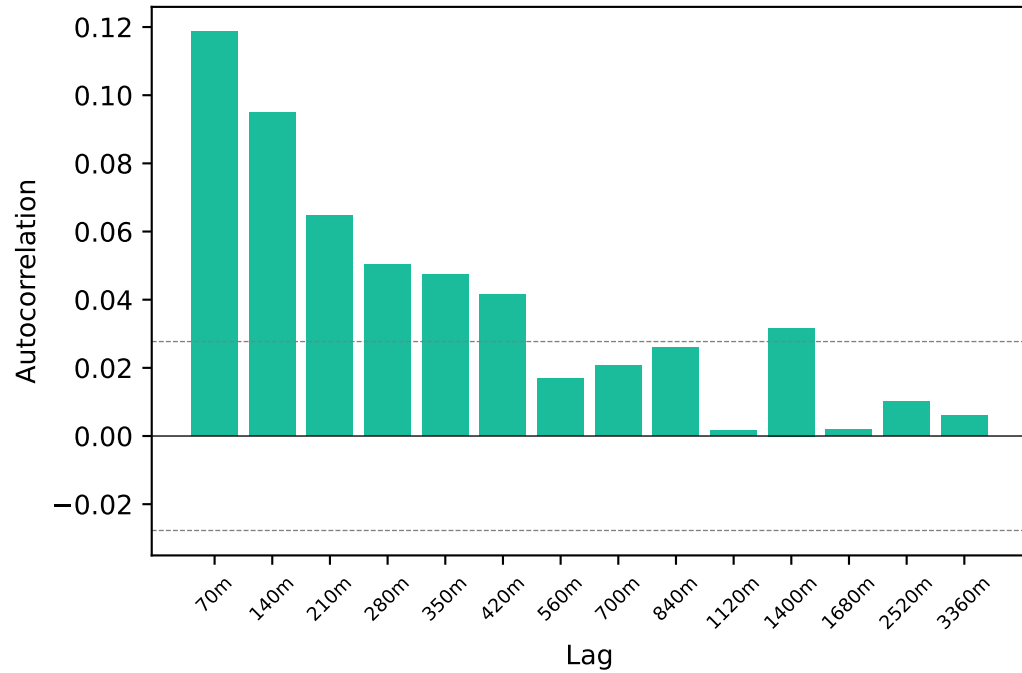


Temporal Clustering: Autocorrelation & Streaks

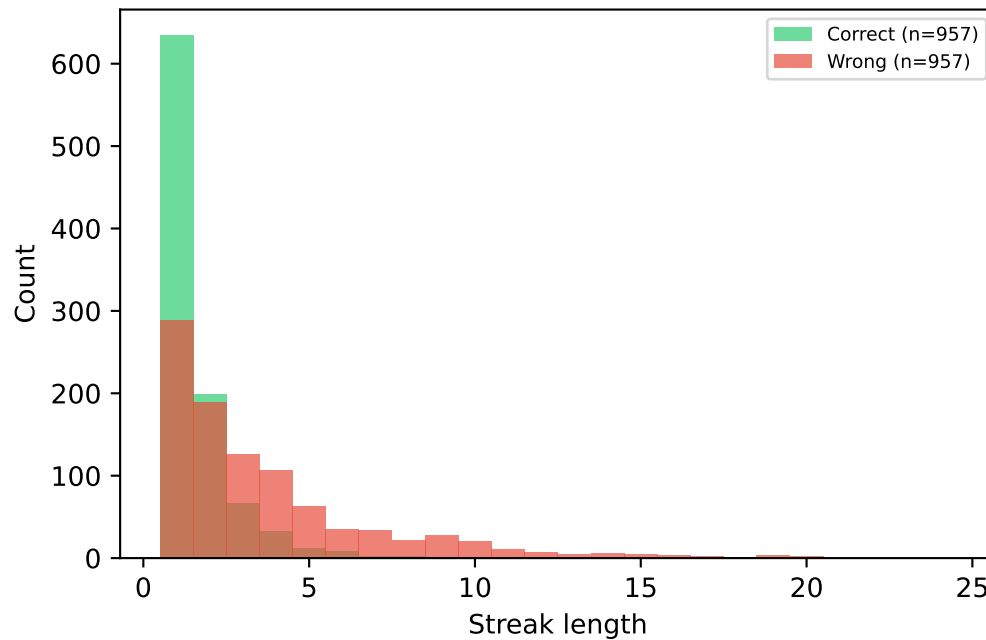
S1 — Autocorrelation



S2 — Autocorrelation



S1 — Streak Distribution



S2 — Streak Distribution

